Differential Amplifiers Experiment 2

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Pre-LAB Work

1. For the circuit given in Fig. 1, calculate the differential mode (Vout/Vid) and common mode (Vout/Vcm) voltage gains. Ignore the Early Effect and assume that all the transistors are operating in active mode. Vcc=15V and Vee=-15V.

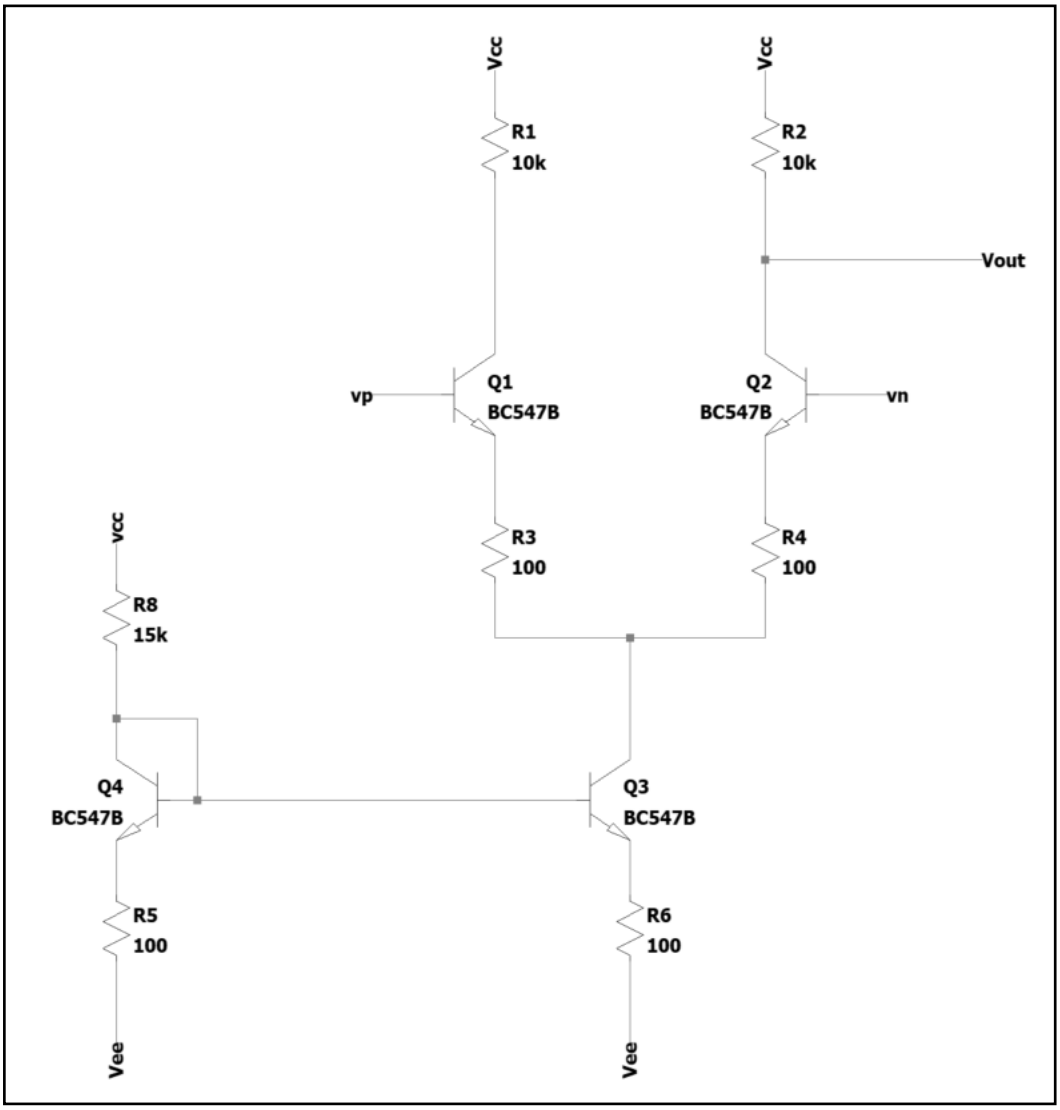
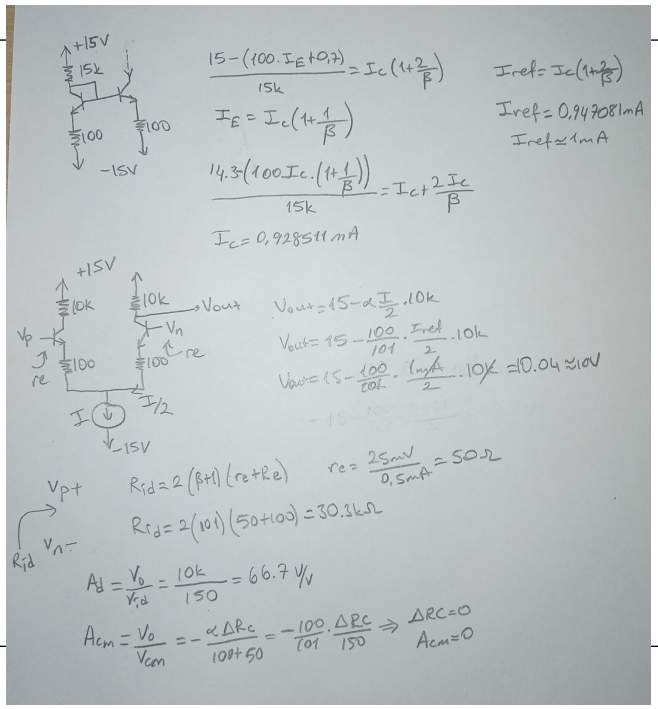


Figure 1

Answer:



2. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Apply a sine wave of 1kHz with an amplitude of 50mV to vp and connect vn to ground. Measure the differential mode voltage gains.

Ad=(Vout[peak-to-peak]/100mV)= 37.767

3. Simulate the circuit in Fig. 1 with SPICE. Let Vcc=15V and Vee=-15V. Apply a sine wave of 1kHz with an amplitude of 50mV to both vp and vn. Measure the common mode voltage gains.

Ac=(Vout[peak-to-peak]/100mV)=0.01548

Equipment/Component List

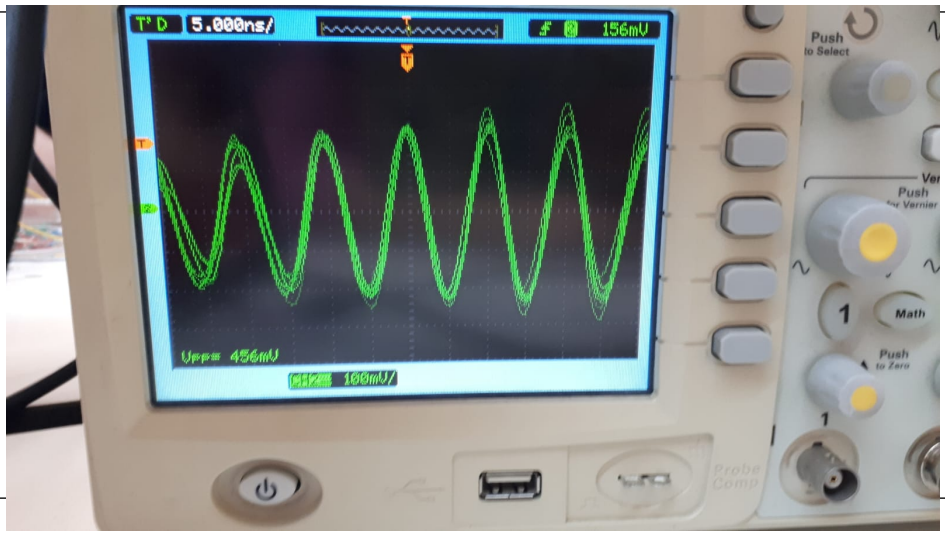
Table 1: Equipment/Component List

Equipment/Component	**Model/Value**	**Count**
Breadboard	-	1
Resistor	100Ω	**4**
Resistor	10kΩ	**2**
Resistor	15kΩ	1
BJT (npn)	BC547	**4**
Multi-meter	-	1
Voltage Source	15V, -15V	1
Probes & Cables	-	-

Experiment Procedure

4. Adjust Vcc to 15V and Vee to -15V.
5. Set-up the circuit in Fig. 1.
6. Connect Vn to ground.
7. Apply a sine wave of 1kHz with an amplitude of 50mV to Vp.
8. Plot Vout and measure the peak-to-peak value of the output.

Plot:



9. Calculate the differential mode voltage gain.

Ad=(Vout[peak-to-peak]/100mV)= 30.5mV

10. Remove the ground connection from Vn and then connect Vn to Vp.
11. Calculate the common mode voltage gain.

Ac=(Vout[peak-to-peak]/100mV)= 0.00195mV

12. Compare the calculated (1), simulated (2, 3) and measured (9, 11) common and differential mode voltage gains.

Answer:

In the conclusion, we get some different responses on the circuit when calculated, simulated, and measured. On the lab measurement, we can't get a stable result. We think that some differences caused by the equipment are not stable due to environmental effects. In addition, our calculation results and simulation results do not differ too much.

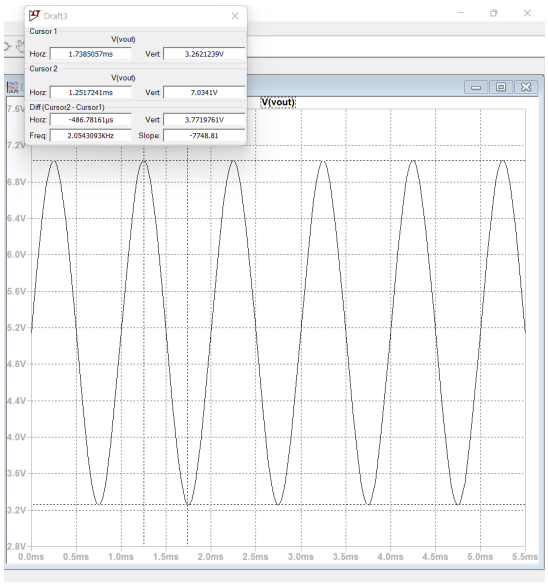


Figure 1: Ltspice Simulation result when Vn is connected to the ground

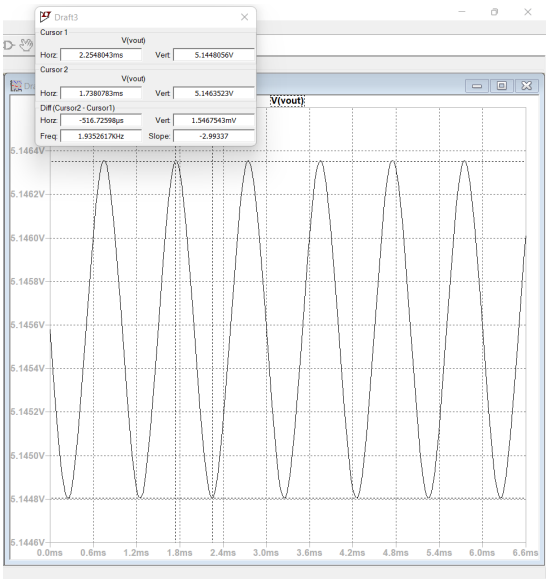


Figure 2: Ltspice Simulation result when Vn is connected to the -15V